🔺 Counting Figures — Triangles, Squares & Cubes (SSC Tricks)

The non-verbal "count the figures" questions — with formulas, figures & shortcuts. Triangles, squares, rectangles, and painted-cube problems.

Golden rule: count *simplest figures first*, then combinations of 2, 3, … Be systematic — label parts and add up. Use the formulas where the figure is a standard grid.

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🧭 The systematic counting method ↑ top

Don't count randomly — you will miss some. Steps:
1. Count the **smallest** (single-part) figures and label them 1, 2, 3…
2. Count figures made of **2 parts**, then **3 parts**, and so on.
3. Add all levels. For standard grids/fans, use the **formulas** below instead.

Counting Triangles ↑ top

(a) Big triangle divided by lines from the apex to the base. If the base is divided into **n** parts (n cevians create n small triangles):

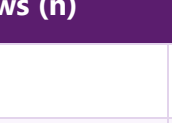


```
Base split into n segments by lines from apex.
Number of triangles = n + (n-1) + ... + 1
                    = n(n+1)/2

e.g. n = 3  →  3·4/2 = 6 triangles
```

(b) Triangle made of a grid of rows (like a triangular wall of small triangles). For a big triangle divided into smaller ones with **n** rows:

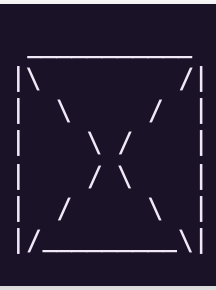
Rows (n)	Total triangles	Rule
1	1	If n is even → `n(n+2)(2n+1)/8` If n is odd → `[n(n+2)(2n+1) − 1]/8`
2	5	
3	13	
4	27	
5	48	



```
n = 2 rows

Count: 4 small (up) + 1 inverted + ...
small upward: 1+3 = 4
inverted: 1
TOTAL = 5  ✓ (matches table)
```

(c) Square with both diagonals (very common).



```
A square + both diagonals
= 8 triangles
  (4 small + 4 half-square)

If it also has the mid-lines (a "Union-Jack"),
count grows — go level by level.
```

Counting Squares ↑ top

In an **n × n** grid (a chessboard-type square divided into unit squares):

```
3 × 3 grid:          Squares of size 1×1 : 9
 _ _ _               Squares of size 2×2 : 4
|_|_|_|              Squares of size 3×3 : 1
|_|_|_|              TOTAL = 9 + 4 + 1 = 14
|_|_|_|              = 1² + 2² + 3²
```

Grid	Total squares	Formula
2 × 2	5	$n^2 + (n-1)^2 + \dots + 1^2$ = $n(n+1)(2n+1)/6$
3 × 3	14	
4 × 4	30	
8 × 8 (chessboard)	**204**	

▭ Counting Rectangles ↑ top

In an **m × n** grid, total rectangles (squares included):

```
Total rectangles = [m(m+1)/2] × [n(n+1)/2]
(choose 2 vertical lines × 2 horizontal lines)

e.g. 3 × 2 grid  =  (3·4/2) × (2·3/2) = 6 × 3 = 18 rectangles
Chessboard 8×8   =  (8·9/2)² = 36² = 1296 rectangles
```

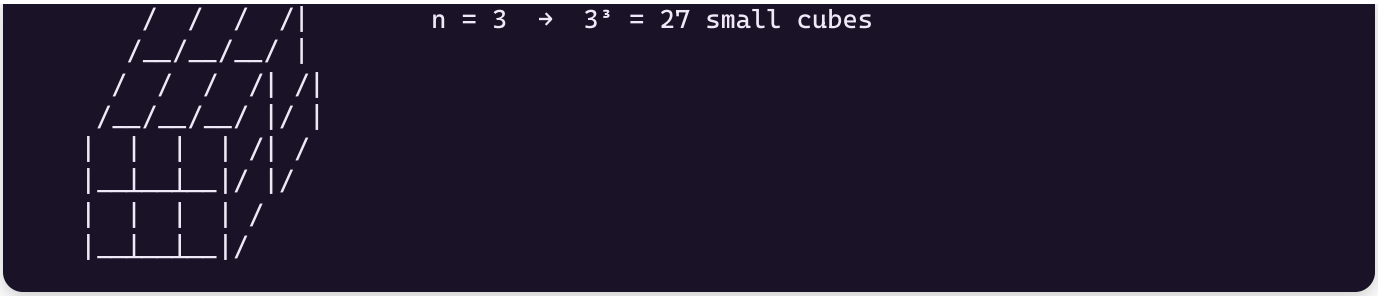
Grid	Rectangles (incl. squares)
2 × 2	9
3 × 3	36
8 × 8	1296

Only rectangles that are NOT squares = total rectangles − total squares.

 Cube cut into smaller cubes ↑ top

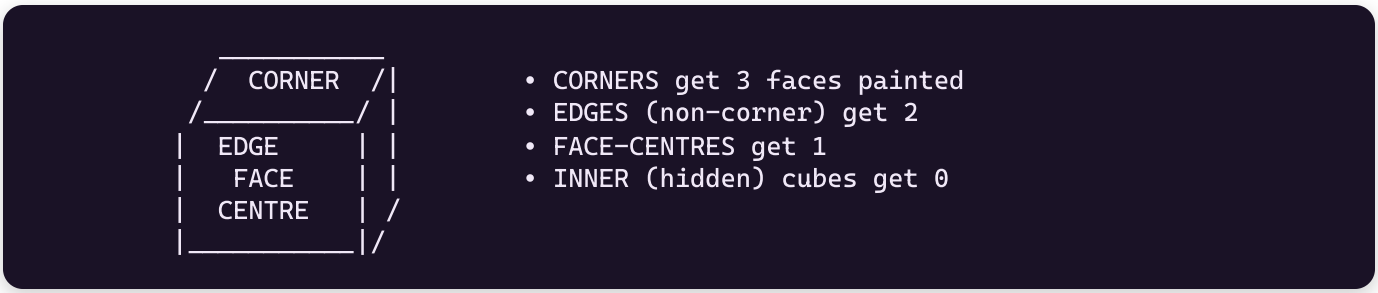
```
A cube cut n times along each of the 3 directions
→ n × n × n = n³ small cubes.

      _______
```



🎨 Painted Cube problems (SSC favourite) ↑ top

> A cube is painted on all 6 faces, then cut into **n^3** small cubes. How many small cubes have paint on 3, 2, 1, or 0 faces? **Memorise these 4 formulas.**



Paint on…	Where	Formula	n = 3	n = 4
3 faces	the 8 corners	`8` (always)	8	8
2 faces	along the 12 edges	`12 × (n − 2)`	12	24
1 face	centre of the 6 faces	`6 × (n − 2)²`	6	24
0 faces	hidden inside	`(n − 2)³`	1	8
Check: total		`= n³`	27	64

□ **Sanity check:** $8 + 12(n-2) + 6(n-2)^2 + (n-2)^3$ must equal n^3. For n = 4: 8 + 24 + 24 + 8 = **64 = 4^3** ✓

⚠ **Special cases SSC asks:** if only *some* faces are painted, or two opposite faces left unpainted, recompute — the corner/edge counts change. Also: *at least one face painted* = $n^3 - (n-2)^3$.

⚡ Quick formulas (cover the right) ↑ top

Figure	Formula
Triangle, apex lines, base in n parts	$n(n+1)/2$
Squares in an n×n grid	$1^2+2^2+\ldots+n^2 = n(n+1)(2n+1)/6$
Rectangles in m×n grid	$[m(m+1)/2]\cdot[n(n+1)/2]$
Square + both diagonals → triangles	8
Cube cut → small cubes	n^3
Painted cube — 3 faces	8 (corners)
Painted cube — 2 faces	$12(n-2)$
Painted cube — 1 face	$6(n-2)^2$
Painted cube — 0 faces	$(n-2)^3$
At least one face painted	$n^3 - (n-2)^3$

Exam tip: for irregular figures (no formula), *always* count level by level (1-part, 2-part, …) and label — that's how toppers avoid missing figures. For grids and painted cubes, the formulas above give the answer in seconds.